

Quantum Electronics & Qubits

Exercise Sheet 6

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Exercise 1: All about Quantum Dots

Part A: Electron transfer process

Consider a single-level quantum dot (QD) of capacitance C_Σ coupled to the source S and drain D via identical tunnel barriers. No further electrodes are present, hence $-C_\Sigma = C_{\text{QD-S}} + C_{\text{QD-D}} = 2C_{\text{QD-S}}$. We neglect the size quantization and assume that transport occurs via the ground state of the QD. Furthermore, there are no background charges. A voltage V is applied to S with respect to the grounded D.

1. Draw the energy diagram of the system for $V = 0$. At what temperature is the charge quantization at the QD smeared out?
2. Draw the energy diagram for $V = e/C_\Sigma$ before any electron transfer. Calculate the energy differences of the 4 charge transfers to be considered. Which of them may occur at $k_B T \ll e^2/(2C_\Sigma)$?
3. Suppose that under the condition of 2., one electron has tunneled from S to the island. Draw the energy diagram for this situation. Calculate the energy differences for all 4 imaginable charge transfers. Which electron transfer is next, and how does the energy diagram look like after this transfer?

Part B: Current through a Quantum Dot

Suppose the energy level of the dot is ε_0 and the chemical potentials of the source (left) and drain (right) leads are μ_L and μ_R with temperatures T_L and T_R respectively. The tunnel couplings are characterized by tunnel rates Γ_L and Γ_R for the left and right barriers, respectively. The current I through the dot in terms of the Fermi functions and the transmission probability can be written as:

$$I = \frac{e}{h} \int_{-\infty}^{\infty} d\varepsilon T(\varepsilon) [f_L(\varepsilon) - f_R(\varepsilon)]$$

where $T(\varepsilon)$ is the energy-dependent transmission function. For a single-level quantum dot that we are considering here, the transmission function has the Breit-Wigner form:

$$T(\varepsilon) = \frac{\Gamma_L \Gamma_R}{(\varepsilon - \varepsilon_0)^2 + (\Gamma/2)^2} \quad \text{with} \quad \Gamma = \Gamma_L + \Gamma_R,$$

1. Linearize the current in the small bias voltage $V = (\mu_L - \mu_R)/e$ and derive the differential conductance $G = \frac{dI}{dV}|_{V=0}$.
2. Show how the zero-bias conductance G depends on temperature and the position of ε_0 relative to the Fermi level. Sketch $G(\varepsilon_0)$ for several values of temperature.

Part C: Thermometry Using a Quantum Dot

In the regime where $E_C \gg \delta E \gg k_B T$ meaning the charging energy is much larger than the single-particle level spacing of the quantum dot and the thermal energy, the conductance peaks in gate voltage has single periodicity and the amplitude of the peaks are constant. In this so-called classical Coulomb blockade regime with bias voltage close to zero, the conductance through the dot is given by:

$$G/G_\infty = \frac{\delta/k_B T}{2 \sinh(\delta/k_B T)} \approx \frac{1}{2} \cosh^2 \left(\frac{\delta}{2.5 k_B T} \right)$$

where G_∞ is the high-temperature value of the conductance and the δ measures the distance to the center of the conductance peak and has units of energy.

1. Explain qualitatively how the width of the conductance peak G can be used to determine the electron temperature.
2. Derive an expression for the full width at half maximum (FWHM) of the conductance peak in the linear response regime. How does the FWHM scale with temperature?
3. Describe a measurement protocol where one extracts the electron temperature from the shape of the conductance peak. What assumptions must be made for the method to be accurate?
4. Discuss potential systematic errors in this method (e.g., level broadening, noise, nonlinearity). How could these be minimized in an experiment?

See Iftikhar, Z., *et al.* Primary thermometry triad at 6 mK in mesoscopic circuits. *Nat Commun* 7, 12908 (2016) for an example of the electron temperature measurement using three different methods one of which is the Coulomb blockade thermometry.

Exercise 3: Quantum capacitance of Graphene

We will have a look at the study by Xia *et al.* in Nature Nanotechnology 4, 505-509 (2009).

1. Explain the concept of quantum capacitance. How does it differ from classical capacitance? Why is quantum capacitance particularly significant in low-dimensional materials like graphene?
2. Describe the experimental setup used by Xia *et al.* to measure the quantum capacitance of graphene, see Fig. 1(a). What role does the ionic liquid play in their measurement?
3. The geometrical capacitance stemming from the ionic double layer can be modelled by a parallel plate capacitor for which the capacitance per unit area is given by $\epsilon_0 \epsilon_r / t$, where $\epsilon_0 = 8.85 \times 10^{-12} \text{ Fm}^{-1}$, $\epsilon_r \approx 7$ is the dielectric constant of the ionic liquid, and $t \approx 0.3 \text{ nm}$ is the radius of the counter-ions. Explain how the quantum capacitance can be extracted from the measured total capacitance in Fig. 1(b).
4. Write the quantum capacitance of graphene as a function of the Fermi energy $E_F = eV$. How does C_q behave as a function of V ? Compare this behaviour with Fig. 1(b). Can you explain the discrepancy?
5. Can you think of some of the challenges associated with measuring the quantum capacitance of graphene using an ionic liquid gate? How might these challenges affect the accuracy and reliability of the measurements?

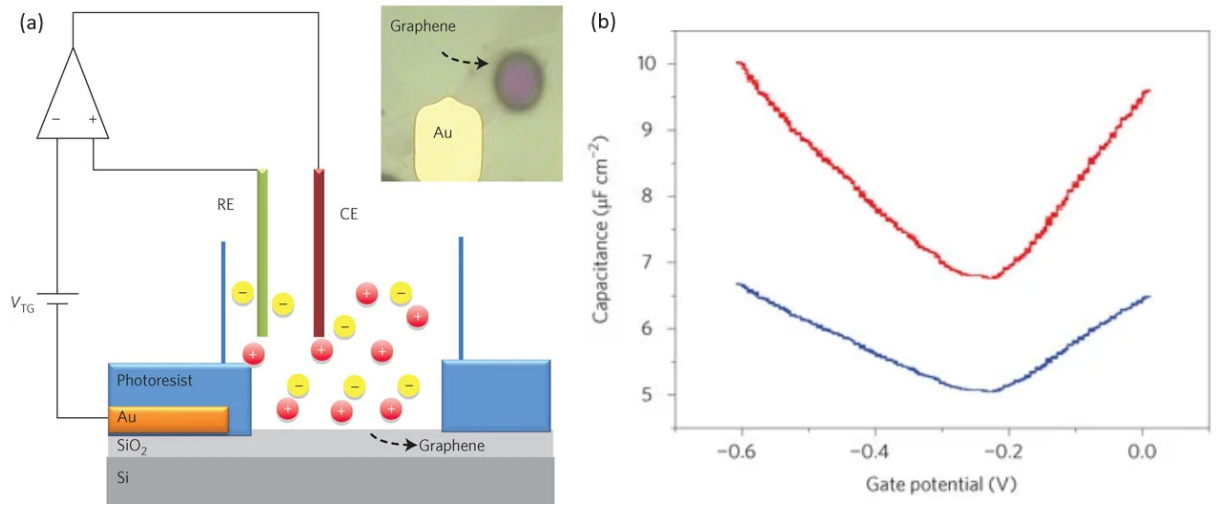


Figure 1: (a) Schematic of the quantum capacitance measurement setup in which a graphene sheet on a silicon/SiO₂ substrate is connected to a gold electrode. The edge of the graphene sheet is covered with an insulation layer so that only the top surface is exposed to an ionic liquid electrolyte. The potential of the graphene is controlled and varied with respect to a platinum reference electrode (RE) using a three-electrode electrochemical configuration, with an operational amplifier (opamp) acting as a comparator. V_{TG} is the electrochemical gate voltage, and CE denotes the counter electrode. The inset is an optical micrograph of the graphene device. (b) Total capacitance (blue line) and quantum capacitance (red line) of graphene measured in ionic liquid BMIM-PF₆. The potential is measured with respect to a platinum quasi-reference electrode (RE).